

LALITHA RAO

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EDUCATION

The Cooper Union for the Advancement of Science and Art

New York, NY

Bachelor of Engineering in Mechanical Engineering

2024 – 2028

- **Relevant Coursework:** Thermodynamics, Feedback & Control, Fluid Mechanics, Computer Aided Engineering

WORK EXPERIENCE

Machines in Motion Laboratory (NYU Tandon)

New York, NY

Research Intern

June 2026 – Present

- Engineered a 9 DoF wearable data-collection glove adapting a Pollen Robotics chassis with custom AS5600 encoders and Intel RealSense T265 cameras.
- Streamed synchronised 9 channel joint angles and spatial tracking telemetry directly into a MuJoCo physics simulation in real time for kinematic verification with minimal latency.
- Processed 3 hours of trajectory datasets with Python and PyTorch (LeRobot) to format data for imitation learning models.

The Cooper Union

New York, NY

STEM Ambassador

2024 – 2025

- Conducted hands-on STEM workshops and demonstrations for New York City high school students.

RESEARCH EXPERIENCE

Aero-Hydro Coupling in Multimodal Insects

2025 – Present

- Designed and conducted experiments validating theoretical power output models in flapping locomotion.
- 3D-printed whirligig beetle wing models with varying morphological parameters.
- Built an Arduino-controlled flapping mechanism with adjustable frequency and amplitude.

Surface Tension of Water: Effects of pH, Salt Concentration, and Temperature

2023 – 2024

- Measured surface tension using capillary rise and Tate's Law methods.
- Compared experimental results with theoretical values and performed statistical analysis.

ENGINEERING PROJECTS

ME 211 Lifting System

- Designed and 3D printed a gear train using spur gears with a gear ratio of 1:6. Decided gear ratio based on motor stall torque and load calculations.
- Enabled a NEMA 17 stepper motor to lift a 5 lb weight 2 ft in 16 s.

ME-200 Folk Acrobat Toy Dynamic Modeling

- Derived equations of motion and implemented numerical simulations in Python.
- Validated dynamic predictions using optical motion tracking and statistical error analysis.

EID 102 Autonomous Robot Design

- Designed and programmed an autonomous multi-task robot implementing closed-loop PID line-following.
- Derived rotational kinematics ($s = r\theta$) and sensor-offset geometry to execute precise 90° turns and ultrasonic braking at 300 mm wall proximity.

WRITING & OUTREACH

Author, *The Flow of Light* (Published on Amazon)

2022 – 2024

- Authored an introductory physics text on optics and light behavior aimed at secondary school students.

Society of Women Engineers (75th Anniversary)

2024

- Researched and wrote biographical profiles on women in STEM for a commemorative historical presentation.

TECHNICAL SKILLS

- **Programming & Software:** Python, PyTorch, LeRobot, MuJoCo, MATLAB, Simulink, Arduino, CAD, $\text{\LaTeX}$
- **Hardware & Prototyping:** 3D Printing (Slicing/FDM), Machining, Sensor Integration, Actuators
- **Engineering Core:** Kinematic Modeling, Control Systems, Dynamic Simulation, Experimental Design